

	PRODUCT SPECIFICATION		Form No.: XXXXXX	
	Author/Dept.: Wang Le/CPD		Document #:	Rev: 1.0
	Production Specification of 324Ah Cell			
	Confidential: () Level 3 () Level 2 (V) Level 1			

Product Specification

Model: QKL0M5

Prepared by CPD	Approved by CPD	Approved by SLS	Approved by EV- PRJ	Approved by QA	Approved by Product Manager
Wang Le					

Customer Approval	Signature	Data
	Customer company code:	
	Customer company stamp:	

Customer Inquiry

Model: QKL0M5

Version: 1.0

The customer is requested to write down your information and contact CATL in advance. If the customer needs some special applications or operating conditions not described in this document, CATL could design and build such products according to the customer's special requirements.

	Special Requirements	Criteria
1		
2		
3		
4		
5		

Company code:

Signature:

Date:

Amendment Records

Rev.	ECN No.	Effective Date	Author	Description of Revision
1.0			Wang Le	New release

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Definitions

Terms	Definition
Product	Means the 324Ah 3.22V rechargeable lithium ion cells produced by CATL.
Customer	Means the buyer in the Contract.
CATL	Means the seller in the Contract.
PN	Means the unique part number provided by CATL to identify the product.
Ambient Temperature	Ambient air temperature of the environment to which the product is exposed.
BMS	Battery system management.
Cell temperature	Means the temperature of product as measured by a thermal sensors.
Fresh State	Means cell state within 7 days after production.
Charge C-Rate	Means the ratio of charging current to the latest cell capacity as frequently measured by the Battery Management System. For example, the initial cell capacity is 324Ah and a Charge C-Rate of 0.2C equals to a charge current of 64.8A. When the cell capacity declined to 260Ah and a Charge C-Rate of 0.2C equals to 52.0A.
Cycle	Means a state when a total of standard charge and standard discharge according to rules from a cell as recorded by BMS and it may consist of a summation of a few segments of partial discharges.
Production date	Means the production date of the cell marking on the top of the cell by date code.
OCV	Means open circuit voltage.
Product supply agreement	Transaction signed by customer and CATL about the product in this specification.
Standard charge	Charge method set out in paragraph 2.2.4.
Standard discharge	Discharge method set out in paragraph 2.3.1.
SOC	The linear scale of charge held by a cell as measured by capacity in Ah. 100% SOC means a cell is fully charged at 324Ah while 0% SOC means a cell is fully discharge down to 0Ah. The SOC should indicate a no load situation.
Temperature rise	Means the increase of cell temperature from one state to another in certain events such as charging or discharging.
Unit of measure	“V” (Volt), voltage unit “A” (Ampere), current unit “Ah” (Ampere-Hour), capacity unit “Wh” (Watt-Hour), energy unit “Ω” (Ohm) , resistor unit “mΩ” (milliohm), resistor unit “°C” (degree Celsius), temperature unit “mm” (millimeter), length unit “s” (second), time unit “Hz” (Hertz), frequency unit

1 Scope

The purpose of this document is to specify the specifications, application condition and risk information of 324Ah 3.22V rechargeable lithium iron phosphate cells to be supplied by CATL.

2 Electrical performance

2.1 General

No.	Parameter	Specification	Condition
2.1.1	Typical capacity	324Ah	Fresh state, 25±2°C, 1C
2.1.2	Min. capacity	324Ah	Fresh state, 25±2°C, 1C
2.1.3	Operation voltage	2.5~3.65V 2.0~3.65V	Temperature >0°C Temperature ≤0°C
2.1.4	Impedance (1kHz)	0.21±0.06mΩ	Fresh state (40%SOC)
2.1.5	Shipping capacity	40%±5%	Fresh state (40%SOC)
2.1.6	Self-discharge rate	≤3.5%/month	25°C, 40%SOC, 3 months after fresh state
2.1.7	Operation temperature (charge)	-5~65°C	See paragraph 2.2
2.1.8	Operation temperature (discharge)	-35~65°C	See paragraph 2.3
2.1.9	Cell weight	5.550±0.2kg	N.A.
2.1.10	Cell dimension	See cell drawing	Under 300kgf clamp force in cell thickness direction, BOL
2.1.11	Cycle life	≥3500cycles	Under initial 300kgf clamp force, 25°C, 1C/1C, fading to 80% of initial capacity

2.2 Charge mode/Parameter

No.	Parameter	Specification	Condition
2.2.1	Standard charge current	0~80%SOC1C; 80%~90%0.8C; 90%~95%0.565C; 90%~100% 0.33C	25±2°C
2.2.2	Maximum continuous charge current	1C	25±2°C, ≤80%SOC
2.2.3	Standard charge voltage	Max. 3.65V	N.A.
2.2.4	Standard charge mode	Charge to 3.65V of the cell according to 2.2.1 step charging method, and then charge to 3.65V until the current drops to 0.05C, 25±2°C.	
2.2.5	Standard charge temperature	25±2°C	N.A.
2.2.6	Absolute charge temperature	-5~65°C	Stop charge once cell temperature is outside this range regardless of whether continuous or pulse current is adopted.

2.2.7	Absolute charge voltage	Max. 3.65V	Stop charge once cell temperature is outside this range regardless of whether continuous or pulse current is adopted.
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2.2.8 Other charge condition (mode) Unit: C-Rate

T\SOC	0%	10%	20%	30%	40%	50%	60%	70%	80%	85%	90%	100%
-10℃	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
-5℃	0.20	0.20	0.10	0.10	0.10	0.06	0.06	0.06	0.06	0.04	0.02	0.01
0℃	0.50	0.50	0.30	0.30	0.20	0.20	0.20	0.12	0.12	0.10	0.10	0.04
5℃	0.70	0.70	0.70	0.50	0.40	0.40	0.40	0.20	0.20	0.12	0.12	0.08
10℃	0.80	0.80	0.80	0.80	0.60	0.60	0.60	0.40	0.40	0.30	0.30	0.12
15℃	0.90	0.90	0.90	0.90	0.90	0.90	0.80	0.60	0.60	0.40	0.40	0.20
20℃	1.00	1.00	1.00	1.00	1.00	1.00	0.90	0.90	0.80	0.80	0.60	0.27
25℃	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.80	0.80	0.33
30℃	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.80	0.80	0.33
35℃	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.80	0.80	0.33
45℃	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.80	0.80	0.33
50℃	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.80	0.80	0.33
55℃	0.80	0.80	0.80	0.80	0.80	0.80	0.80	0.80	0.80	0.64	0.64	0.33
60℃	0.28	0.28	0.28	0.28	0.28	0.28	0.28	0.28	0.28	0.22	0.22	0.22
65℃	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

2.3 Discharging mode/Parameter

No.	Parameter	Specification	Condition
2.3.1	Standard discharge current	1/3C	25±2℃
2.3.2	Maximum discharge current (continuous)	1C	N.A.
2.3.3	Discharge cut-off voltage	2.5V 2.0V	Temperature >0℃ Temperature ≤0℃
2.3.4	Standard discharge temperature	25±2℃	N.A.
2.3.5	Absolute discharge temperature	-35~65℃	Stop discharging once cell temperature is outside this range regardless of whether continuous or pulse current is adopted

2.4 Pulse discharge & charge mode

2.4.1 Pulse discharge mode Unit: C-Rate

30s Pulse discharge rate/Cell level													
T\SOC	0.0	5.0	10.0	20.0	30.0	40.0	50.0	60.0	70.0	80.0	90.0	95.0	100.0
-35°C	0.00	0.00	0.0	0.0	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1
-30°C	0.00	0.03	0.06	0.13	0.25	0.25	0.25	0.25	0.25	0.25	0.25	0.25	0.25
-25°C	0.00	0.06	0.13	0.25	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50
-15°C	0.00	0.06	0.13	0.25	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50
-10°C	0.00	0.19	0.38	0.63	1.20	1.20	1.20	1.20	1.20	1.20	1.20	1.20	1.20
-5°C	0.00	0.25	0.50	1.00	2.13	2.13	2.13	2.13	2.13	2.13	2.13	2.13	2.13
0°C	0.00	0.28	0.56	1.06	2.18	2.18	2.18	2.18	2.18	2.18	2.18	2.18	2.37
5°C	0.00	0.31	0.63	1.13	2.25	2.25	2.25	2.25	2.34	2.44	2.54	2.59	2.85
10°C	0.00	0.33	0.66	1.22	2.43	2.46	2.66	2.80	2.93	3.00	3.00	3.00	3.00
15°C	0.00	0.34	0.68	1.32	2.63	2.63	2.78	2.82	2.95	3.00	3.00	3.00	3.00
20°C	0.00	0.36	0.72	1.41	2.82	2.82	2.82	2.85	2.97	3.00	3.00	3.00	3.00
25°C	0.00	0.38	0.75	1.50	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00
30°C	0.00	0.38	0.75	1.50	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00
35°C	0.00	0.38	0.75	1.50	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00
40°C	0.00	0.38	0.75	1.50	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00
45°C	0.00	0.38	0.75	1.50	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00
50°C	0.00	0.38	0.75	1.50	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00
55°C	0.00	0.38	0.75	1.50	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00
60°C	0.00	0.38	0.75	1.50	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00
65°C	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

2.4.2 Pulse charge mode Unit: C-rate

30s Pulse charge rate/Cell level													
T\SOC	0.0	5.0	10.0	20.0	30.0	40.0	50.0	60.0	70.0	80.0	90.0	95.0	100.0
-15°C	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
-10°C	-0.40	-0.40	-0.33	-0.33	-0.30	-0.30	-0.30	-0.20	-0.20	-0.20	-0.14	-0.14	0.00
-5°C	-0.55	-0.55	-0.46	-0.46	-0.39	-0.39	-0.39	-0.26	-0.26	-0.26	-0.20	-0.20	0.00
0°C	-0.72	-0.72	-0.65	-0.65	-0.55	-0.55	-0.46	-0.46	-0.39	-0.39	-0.33	-0.33	0.00
5°C	-0.85	-0.85	-0.85	-0.65	-0.65	-0.65	-0.54	-0.54	-0.54	-0.54	-0.54	-0.54	0.00
10°C	-1.05	-1.05	-1.05	-1.05	-1.05	-1.05	-1.05	-1.05	-1.05	-1.05	-1.05	-1.05	0.00

15°C	-1.22	-1.22	-1.22	-1.22	-1.22	-1.22	-1.22	-1.22	-1.22	-1.22	-1.22	-1.21	0.00
20°C	-1.61	-1.61	-1.61	-1.61	-1.61	-1.61	-1.61	-1.61	-1.61	-1.61	-1.61	-1.22	0.00
25°C	-2.00	-2.00	-2.00	-2.00	-2.00	-2.00	-2.00	-2.00	-2.00	-2.00	-2.00	-1.33	0.00
30°C	-2.00	-2.00	-2.00	-2.00	-2.00	-2.00	-2.00	-2.00	-2.00	-2.00	-2.00	-1.33	0.00
35°C	-2.00	-2.00	-2.00	-2.00	-2.00	-2.00	-2.00	-2.00	-2.00	-2.00	-2.00	-1.33	0.00
40°C	-2.00	-2.00	-2.00	-2.00	-2.00	-2.00	-2.00	-2.00	-2.00	-2.00	-2.00	-1.33	0.00
45°C	-2.00	-2.00	-2.00	-2.00	-2.00	-2.00	-2.00	-2.00	-2.00	-2.00	-2.00	-1.33	0.00
50°C	-2.00	-2.00	-2.00	-2.00	-2.00	-2.00	-2.00	-2.00	-2.00	-2.00	-2.00	-1.33	0.00
55°C	-1.60	-1.60	-1.60	-1.60	-1.60	-1.60	-1.60	-1.60	-1.60	-1.60	-1.60	-1.07	0.00
60°C	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

2.5 Low temperature capacity

No.	Parameter	Specification	Condition
2.5.1	Discharge capacity@0°C	≥85%	Fresh state, 0°C, 1C, 2.0V~3.65V, based on 324Ah
2.5.2	Discharge capacity@-20°C	≥75%	Fresh state, -20°C, 1C, 2.0V~3.65V, based on 324Ah

2.6 Safety and reliability

All test result meet GB/T 38031-2020 standard requirement.

3 Product EOL management

This cell is designed to service with a finite life time. The customer shall develop and implement an active tracking system to monitor and record impedance of each product in its entire service life. CATL and its customer shall come into agreement about internal resistance and capacity measurement methods, CATL and/or its customer shall stop using any of the product when its resistance exceeds 150% of its original internal resistance or its capacity fading to ≤70%(25°C). Failure to comply with this requirement shall render CATL's warranties under the contract inapplicable, thereby releasing CATL from any liability in connection therewith.

4 Application Conditions

Customer shall ensure that the following application conditions in connection with the products are strictly observed:

- 4.1 Customer shall ensure that each cell shall be used under the strict monitor, control and protection by the battery management system.
- 4.2 Customer shall provide detailed information of the BMS, including but not limited to its design, feature, setting, and data file format to CATL for design review and record keeping;
- 4.3 Once that detailed information of the BMS has been reviewed and agreed by CATL, customer shall not modify or change the design, features, setting or data file format of the BMS without the prior written agreement by CATL;
- 4.4 Customer shall keep relevant records of the BMS monitoring data throughout the entire service life of each products, which will be used in the determination and judgment of any product warranty and liability claim

entitlement. No warranty or liability claim will be considered without BMS diagnosis records (at a regular basis, esp. during maintenance) of the relevant product.

4.5 The BMS shall include the following monitoring and control features as a minimum requirement:

No.	Parameter	Specification	Action
4.5.1	Stop charging	3.65V	Stop charge when cell voltage reach to 3.65V.
4.5.2	Over charging protection-level 1	$\geq 3.8V$	Stop charge when cell voltage reach to 3.8V.
4.5.3	Over charging protection-level 2	$> 4.0V$	Lock up BMS until technical trouble shooting when cell voltage reaches 4.0V.
4.5.4	Stop discharging	Min. 2.5V	Stop discharge, minimize the discharging current when cell voltage reach to 2.5V.
4.5.5	Over discharge protection-level 1	Min. 2.0V	Stop discharge, minimum the discharge current when cell voltage reach to 2.0V.
4.5.6	Over discharge protection-level 2	Min. 1.8V	Lock up BMS until technical trouble shooting when cell voltage reach to 1.8V.
4.5.7	Short circuit protection	No short circuit allowed	Disconnect cell terminals by contractor when short circuit occur.
4.5.8	Over current protection	See paragraph 2.3	Control discharge/charge current by BMS within specification.
4.5.9	Over temperature protection	See paragraph 2.2 & 2.3	Stop discharge/charge when cell temperature exceeds specification.
4.5.10	Charge time out limit	Charge complete within 8 hours	Stop charge if charge time exceeds 8h.

Note: The above No.4.5.2, 4.5.3, 4.5.5, 4.5.6 are the warning clause, draw the attention of customers: When the battery reaches any of the terms described in the above, means that the battery has been used beyond the specifications, the customer shall take protective measures on the battery in accordance with the “protection action” and other relevant provisions of this specification. At the same time, the CATL shall not take any responsibility for the damage in connection therewith.

4.6 Prevent over discharge any product down to over discharge state. A product may be permanently damaged internally when the cell voltage is lower than 1.8V and therefore shall be strictly prohibited, failing what CATL’s warranties under the contract shall cease to apply, thereby releasing the CATL from any liability in connection therewith. After discharge cut-off in accordance with paragraph 2.3.3, internal power consumption of the system should be reduced to a minimum to prolong the idle time before recharge. Customer undertakes to educate the users of the products or other parties who may come to handle the product to recharge the cells at minimum time intervals to prevent reaching the over-discharge state.

4.7 When the products are intended to be stored for a prolonged period of time (more than one month), adjust the SOC to around 40%.

4.8 Prevent charging the products at a temperature which is not allowed under the specification hereunder (including standard charge, optional fast charge, emergency charge and regeneration), otherwise unnecessary degradation of the capacity of the product may occur. Follow the specification on minimum charging and regeneration temperature and use the BMS to control it. Charging at temperature lower than the specification hereunder shall render CATL’s warranties under the contract inapplicable, thereby releasing CATL from any liability in connection therewith.

4.9 The design of Pack must fully consider the heat dissipation of cell. CATL does not take the responsibility due to the overheating of the cell or batteries caused by the thermal design problem of the Pack.

- 4.10 The design of Pack must fully consider the waterproof and dustproof of the cells. The Pack must meet the waterproof and dustproof grade stipulated by the relevant national standards. The CATL does not take the responsibility due to damage to the cell (such as corrosion, rust, etc.) caused by water and dust.
- 4.11 Mixture use of cells with different power PN is forbidden, otherwise, CATL is not responsible for quality assurance.

5 Safety Precautions

Customer shall ensure that the following safety precautions in connection with the product are strictly observed:

- 5.1 Do not immerse cell into water;
- 5.2 Do not drop cell into fire or expose them to any high temperature environment exceeding operation temperature as set out in paragraphs 2.1.7 and 2.1.8, otherwise fire hazards may present. At all use time, cell temperature should not exceed 65°C, shut down system by BMS when it occurs.
- 5.3 Do not short circuit cell terminals, otherwise high current and temperature may cause body injury or fire hazards. Metallic cell terminals are exposed from plastic packaging and safety precautions should be implemented to avoid short circuit during system integration or connection.
- 5.4 Always connect cell terminals according to its labels in right polarity, reverse charging is strictly prohibited.
- 5.5 It is extremely dangerous to overcharge cell which may cause overheating and fire hazards. Multiple level of fail over charge protection should be implemented by hardware and software. See paragraph 4.5.3 and 5.11 for minimum requirement to be adopted by the BMS for protection.
- 5.6 Normal charging should finish within a charging time out limit as set out in paragraph 4.5.9. When charging continuous longer than charging time out limit, it tends to overheat the cell which may cause overheating and fire hazards. A timer should be implemented in the charger circuit and set up properly. In case charging does not terminate normally within charging time out limit, ensure that timer will intervene and stop the charging. See paragraph 5.11.
- 5.7 Product should be securely fixed to solid platform, and power cables should be securely attached by fastener to avoid intermittent contact which may cause arcing and sparks.
- 5.8 It is strictly forbidden to use plastic for package and electrical connection. Improper electrical connection within a cell may cause overheating in service.
- 5.9 In the event of electrolyte leakage, avoid contacting electrolyte with skin or eyes. In case come into contact, wash affected area with large amount of water and seek medical help. Do not swallow any parts or substances within a cell.
- 5.10 Protect cells from mechanical shock, impact and pressure. Internal electrical circuit may short circuit to generate high temperature and fire hazards.
- 5.11 When cell charging is terminated improperly for reasons such as exceeding allowable charging time, cut off due to exceeding charging voltage or cut off due to exceeding charging current, all these events are defined as “improper charge termination”. Such event may indicate that there is current leaking within a cell system or some components have started to malfunction and subsequent charging of such cell system without finding and fixing root cause of problem may cause potential over heat or fire hazards. When such event occurs, the BMS should lock itself up to prevent subsequent charging and notice should be given to the user to return the vehicle to dealer for servicing. Subsequent charging should only be resumed after the system has been thoroughly checked by qualified technician who can identify and fix root cause attributed to the “improper charge termination”.
- 5.12 Performing tests described in paragraph 2.6 may result in fire and explosion of the product. Such test shall only be performed in qualified laboratories by qualified personnel with proper safety precautions taken. Performing these tests in an improper way result in severe personal body injury or property damages.

6 Disclaimer

CATL will not take any responsibility if the cell is not used in accordance with the specification. The damage which cause social impact and affect the reputation of CATL, the customer required to provide compensation to CATL. CATL reserve the right to make corrections, enhancements, improvements and other changes to its products and services per latest issue, and to discontinue any product or service per latest issue. Customer should obtain the latest relevant information before placing orders and should verify that such information is current and complete.

7 Hazard Warning

7.1 Warning Statement

Caution

Battery is potentially dangerous and must be properly protected during operation and maintenance!
Improper operation of the test described in paragraph 2.6 may result in serious personal injury and property damage!
Proper tools and protective equipment must be used to operate the battery!
Battery maintenance must be performed by a professional and safety trained person!
Failure to comply with the above warning may result in serious consequence!

7.2 Hazards Types

Customer acknowledges the following potential hazards in connection with the usage and handling of the product:

- 7.2.1 Working with battery can expose the handler to chemical, shock and/or arcing hazards. Although a person's body might react to contact with direct current voltage differently than from contact with alternate current voltage. Customer shall take a conservative position and consider the risk of shock or electrocution to be the same for both alternate current and direct current exposures greater than 50V.
- 7.2.2 Cells expose its handler to chemical hazards associated with electrolyte used in the cell.
- 7.2.3 When selecting work practices and personal protective equipment, customer and its employees shall consider potential exposure to these hazards and therefore prevent accidental short circuit that can result in electrical arcing, explosion, and/or "thermal runaway" of the cells.

8 Cell Mechanical Drawing

